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DATA DESCRIPTOR

EEG-based brain-computer interface (BCI) dataset for directional word recognition

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We present an EEG dataset recorded from 22 neurologically healthy volunteers (12 native Russian speakers and 10 native Spanish speakers) during overt and covert articulation of six spatial-direction words. Monopolar EEG signals were acquired from 38 electrodes positioned according to the international 10–10 system using a Neurovisor-BMM-52 (NVX) amplifier at 500 Hz. In a subset of participants, electromyography (EMG) was simultaneously recorded from the masseter muscle and laryngeal region to exploratorily characterize articulatory muscle activation. Exploratory spectral and coherence analyses, together with classification using standard machine learning methods (Random Forest, SVM, LDA), confirm the presence of condition-specific neural activity distinguishable by standard classifiers (best accuracy $78 \pm 4\%$). The dataset is intended to support the development and benchmarking of algorithms for inner speech recognition in brain–computer interface applications.

Background & Summary

Brain–computer interfaces (BCIs) remain one of the most promising technologies in neuroscience and rehabilitation medicine, offering new avenues for communication and control of external devices for individuals with severe motor and speech impairments^{1,2}. This is especially critical for patients suffering from locked-in syndrome, Broca's and Wernicke's aphasia³, or amyotrophic lateral sclerosis (ALS), conditions in which the loss of verbal and motor functions is not accompanied by significant cognitive deficits. In this context, the development of non-invasive, ergonomic, and effective BCIs constitutes not merely a scientific challenge but a social imperative. Electroencephalography (EEG) appears particularly promising in this regard, owing to its high temporal resolution, relative ease of acquisition, and suitability for everyday use⁴. Contemporary BCI systems developed for these purposes rely on various neurophysiological phenomena, such as event-related potentials⁵—specifically steady-state visual evoked potentials (SSVEPs) and the P300 component—or motor imagery-related potentials^{6,7}, as well as machine learning approaches for EEG signal classification⁸. However, these approaches often require external stimuli, are frequently incapable of real-time operation, and tend to induce user fatigue, thereby limiting their practical applicability and suitability for prolonged use.

From an evolutionary perspective, speech—particularly inner speech—represents a natural, efficient, and highly developed channel for communication and control. In this study, the term inner speech refers to the cognitive process of mentally articulating words without overt vocalization or visible articulatory movements⁹. This term is used consistently throughout the article to denote this process. Unlike other paradigms, it enables stimulus-independent control, meaning that commands can be generated solely through spontaneous cognitive activity, thereby rendering BCI interaction closer to natural human communication¹⁰. This makes inner speech especially promising for developing neurocommunication systems tailored to patients who have lost both speech and general motor function, enabling intuitive and user-friendly control. Nevertheless, the advancement of BCI systems based on inner speech faces a critical obstacle: the lack of publicly available, standardized, and high-quality datasets. In contrast to paradigms such as P300 or motor imagery, for which public repositories exist (e.g., BCI Competition, OpenNeuro¹¹), there is a severe scarcity of open-access EEG datasets associated with inner speech. This significantly impedes progress in developing and benchmarking pattern recognition and classification algorithms, hampers reproducibility of scientific findings, and restricts access for a broad community of researchers and developers.

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This gap motivated the creation of an experimental dataset comprising EEG patterns associated with inner speech production. The database includes recordings obtained from neurologically healthy volunteers who performed tasks involving both overt and covert articulation of six lexical items denoting spatial directions: “up,” “down,” “left,” “right,” “forward,” and “backward.”; Russian-speaking participants additionally performed a seventh command (“next”), which was not included in the Spanish protocol.

The development of this dataset is justified by several key factors. First, it is grounded in a stimulus-independent paradigm, rendering it especially valuable for the development of autonomous BCIs¹⁰. Second, it includes both overt and covert articulation of the same lexical items, enabling comparative analysis of the associated brain activity patterns^{12,13} and facilitating assessment of the discriminative power of various EEG features for classification purposes. Third, the dataset fulfils contemporary standards for open scientific data, ensuring reproducibility and facilitating adoption by the broader research community.

The primary objective of this dataset is to provide the scientific community with a standardized, well-documented, and readily accessible resource for developing, testing, and benchmarking algorithms for recognizing EEG patterns associated with inner speech. We anticipate that this resource will attract researchers and developers in the fields of brain–computer interfaces, machine learning, and cognitive neuroscience, ultimately fostering the creation of more effective, accurate, and accessible neurocommunication and neurocontrol systems.

Several open-access EEG datasets for inner and imagined speech have been published in recent years. The Kara One database¹⁴ provides recordings from 14 participants during imagined and overt articulation of 11 English phonemic and single-word prompts, acquired with a 64-channel system at 1000 Hz, but does not include EMG monitoring. The “Thinking Out Loud” dataset¹⁵ offers over 9 hours of EEG data from 10 participants across three paradigms, recorded with a high-density 136-channel system, but is limited to a single language. The FEIS dataset¹⁶ comprises 14-channel recordings from 21 participants during imagined and spoken English phonemes, but its low channel count limits spatial resolution. The ArEEG dataset¹⁷ focuses on Arabic characters rather than directional navigation commands. The present dataset occupies a distinct niche: it provides 38-channel EEG with exploratory simultaneous EMG in a participant subset, paired overt and covert articulation of six spatial navigation commands, and includes two typologically distinct languages — Russian and Spanish — within a single standardized experimental protocol.

Methods

Participants. The study involved 22 right-handed volunteers (12 native Russian speakers and 10 native Spanish speakers), aged 18 to 34 years (mean $\pm$ SD: 23.5 $\pm$ 4.1 years), with no history of neurological disorders or prior brain surgery. The electroencephalographic (EEG) activity of all participants corresponded to age-appropriate normative values. Prior to the experiment, participants were interviewed to screen for potential confounding factors, including recent use of caffeine, nicotine, or medications known to affect EEG activity. All experimental sessions were conducted following an adaptation phase of approximately 10 minutes, during which participants were familiarized with the experimental procedure and trained in task execution.

Ethics statement. The study was carried out in accordance with the ethical standards set forth in the Declaration of Helsinki (1964)¹⁸ and under the protocol approved by the Bioethics Commission of Southern Federal University (Bioethics Committee Report No. 3, dated September 9, 2022). Russian-speaking participants were recruited from among students of Southern Federal University. Spanish-speaking participants were recruited from university students and members of the local Spanish-speaking community. All participants were verbally informed about the structure of the experiment and provided written informed consent prior to participation, including consent for their anonymized data to be made publicly available. Prior to data release, all personal identifying information was removed; only age, sex, and native language were retained as participant metadata. Participants are identified exclusively by numerical codes.

Experimental paradigm. The experiment was conducted in a light- and sound-attenuated, electromagnetically shielded chamber (Faraday cage). Participants sat in a comfortable armchair in a relaxed posture to minimize motor activity. During electrode placement, each participant received standardized instructions describing the experimental protocol. A schematic of the experimental setup is presented in Fig. 1.

For the Russian-language dataset, six spatial-direction words were used: “forward,” “backward,” “right,” “left,” “up,” and “down.” The Spanish-language dataset likewise comprised six directional words: “arriba,” “abajo,” “derecha,” “izquierda,” “adelante,” and “atrás.”

Stimulus words were selected based on semantic equivalence to navigation commands rather than phonetic similarity. Russian and Spanish stimuli differ in syllable count and phonological structure, which should be considered when making direct cross-linguistic comparisons at the signal level (see Table 1). The word “next” (Russian: *далее*) was included only in the Russian-language dataset.

Each participant completed one training session and one main experimental session, lasting approximately 70 minutes in total, including time for electrode application and familiarization with the protocol. The session structure is illustrated in Fig. 2.

Each experimental session began with a one-minute recording of resting-state EEG with eyes open, reflecting a baseline of calm wakefulness, and concluded with an identical one-minute recording under the same conditions, thereby enabling the capture of baseline neuroelectrical activity both before and after task performance. Following the initial baseline recording, participants proceeded to the main experimental phase, which consisted of six consecutive blocks—one for each of the six spatial-direction words: “up,” “down,” “left,” “right,” “forward,” and “backward.”



Fig. 1 Experimental setup.

Direction	Russian	Syllables (RU)	IPA (Russian)	Spanish	Syllables (ES)	IPA (Spanish)
Up	Вверх	1	[v'ɛrx]	Arriba	3	[a'riβa]
Down	Вниз	1	[vn'is]	Abajo	3	[a'βaxo]
Right	Вправо	2	[v'pravə]	Derecha	3	[de'retʃa]
Left	Влево	2	[vl'evə]	Izquierda	4	[is'kjerda]
Forward	Вперёд	2	[fp'i'rot]	Adelante	4	[ade'lante]
Back	Назад	2	[na'zat]	Atrás	2	[a'tras]
Next	Дальше	2	[dal'ʃə]	—	—	—

Table 1. Stimulus words used in the Russian and Spanish datasets, with IPA transcription and syllable count.

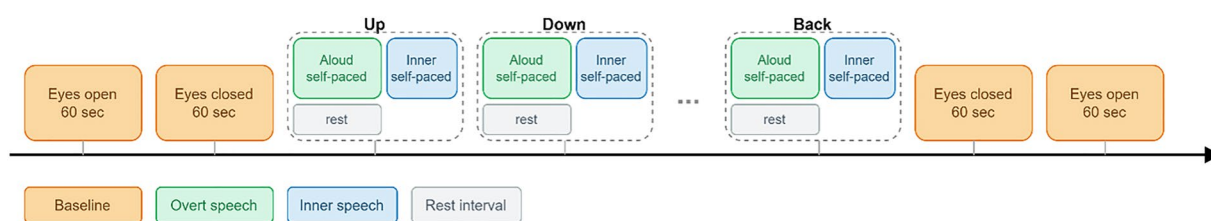


Fig. 2 Schematic of the examination procedure.

Within each block, participants first performed an overt articulation task, during which they clearly and calmly vocalized the target word aloud at a self-paced rate, pressing a designated key on a keyboard immediately before each vocalization to ensure precise synchronization between behavioral events and the EEG recording. The key was pressed using the right hand across all participants and both experimental conditions. Upon completion of this block, a short rest interval was taken at the participant's discretion (mean inter-block gap: 0.4–1.3 minutes), after which a second block commenced. In this second block, participants covertly (i.e., without articulation or vocalization) rehearsed the same word at the same self-paced rate, again marking each inner articulation by pressing the key. The mean duration of individual word blocks ranged from 3.0 to 6.4 minutes across participants (grand mean approximately 4.4 minutes). The inter-trial interval varied across participants and blocks, with a mean of approximately 1.6–7.1 seconds as derived from the recorded event timestamps.

It should be noted that a subset of Russian-speaking participants ($n = 4$; sub1, sub3, sub5, sub10) employed a modified marker protocol during data collection. Rather than placing an event marker immediately before each individual utterance, these participants placed a single marker at the onset of each utterance series, vocalising or covertly rehearsing the target word multiple times without duplicating the marker. For these participants, epochs were extracted at 1-second intervals following each marker for a period of 10–15 seconds. The epoch structure

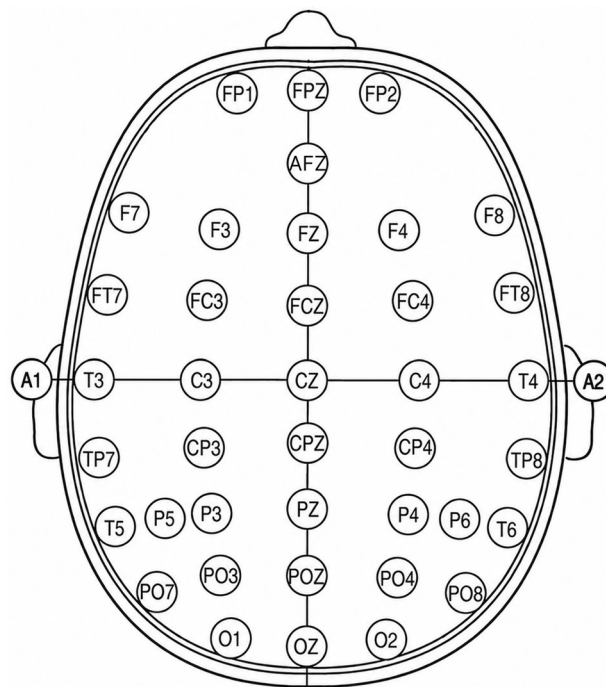


Fig. 3 The international “10-10” electrode placement system. A combined auricular reference (A1 and A2) and ground electrode (AFz) were used.

for these participants therefore differs from the standard single-trial protocol described above, and users of the dataset should take this into account when performing trial-level analyses or making cross-participant comparisons. These participants are listed separately in Table 4 of the Data Records section.

For participants following the standard marker protocol, the number of recorded trials varied across language groups. Russian-speaking participants yielded 672–1306 trials per session (mean per word: 48–93), with one participant (sub6) yielding a lower trial count of 358 due to a shortened recording session. Spanish-speaking participants yielded 1041–2148 trials per session (mean per word: 87–179), reflecting longer or more densely sampled sessions. Trial counts for participants using the modified marker protocol are reported separately in Table 4.

Data collection. Bioelectrical activity was recorded monopolarly (i.e., each electrode referenced to a common combined auricular reference at A1 and A2) from 38 EEG and 2 EMG channels using the “Neurovisor-BMM-52 (NVX)” electroencephalograph manufactured by Medical Computer Systems (Zelenograd, Russia). Flat active electrodes filled with conductive gel were used for signal acquisition. EEG electrodes were positioned on the scalp according to the international 10–10 system¹⁹. EMG was recorded bilaterally from the masseter muscle and from the laryngeal region corresponding to the vocal folds; electrodes were affixed to the skin using medical-grade hypoallergenic adhesive patches. EMG recordings were obtained in a subset of participants ($n = 6$; 2 Russian-speaking and 4 Spanish-speaking) for exploratory characterization of articulatory muscle activation patterns during overt and covert speech. No EMG-based trial rejection was applied.

Prior to recording, electrode impedances were checked using the built-in impedance monitoring of the NVX-52 amplifier and kept at acceptable levels for all participants.

EEG signals were sampled at 500 Hz per channel. A bandpass filter restricted the recorded EEG frequency range to 1–70 Hz. A 50 Hz notch filter was applied to eliminate power-line interference. Reference electrodes were placed on the earlobes, and the ground electrode was positioned at AFz. The complete electrode montage (Fig. 3) enabled the recording of bioelectrical activity from all neocortical structures implicated in various aspects of speech production and processing^{3,13}.

Upon completion of each recording session, data were saved in the standard.edf (European Data Format) file format on a computer, containing continuous recordings from 38 EEG channels, 2 EMG channels, and synchronized event markers.

Brain-computer interface (BCI) interaction conditions. The resulting database was designed to support both research into the mechanisms of inner speech⁹ and the identification of informative EEG features applicable in brain–computer interface (BCI) systems^{10,12}. Participants were instructed to silently articulate words until stopped by the experimenter. To minimize artifacts, they were asked to produce words calmly, reducing associated motor activity as much as possible²⁰. Although motor activity was primarily expected during overt speech, it is well established that inner speech can also be accompanied by muscular activation in articulatory organs, even in the absence of overt movement. Such covert articulatory activity—subvocalization—refers to weak activation

of speech-related muscles during inner speech, detectable via surface electromyography (EMG). It is considered a peripheral marker of phonetic-motor planning during internal speech processing¹³.

In this study, EMG was recorded in a subset of participants to exploratorily assess the degree of articulatory muscle activation during inner speech. The data confirms that inner speech production was accompanied by low-level but detectable muscle activity in the masseter and laryngeal regions, consistent with subvocalization reported in the literature.

Data Processing. To transform the raw electrophysiological recordings into a more compact and analytically tractable dataset, a preprocessing pipeline was employed. Preprocessing was performed offline and implemented in Python using the MNE library¹¹.

During the initial preprocessing stage, raw data were band-pass filtered using a fourth-order Butterworth filter with cutoff frequencies of 1 Hz and 70 Hz. This range was selected to preserve the principal neurophysiological oscillatory bands—including theta, alpha, beta, and low-gamma activity—while simultaneously attenuating slow drifts (<1 Hz) and high-frequency noise (>70 Hz), such as electromyographic artifacts and broadband interference. Subsequently, to specifically suppress power-line interference at the fundamental 50 Hz harmonic, a second-order Butterworth notch filter with a narrow stopband (48–52 Hz) was applied. This effectively eliminated line noise without distorting adjacent frequency components.

Prior to epoching, the continuous data were visually inspected and segments containing prominent artifacts—such as large amplitude transients, channel-level drifts, or excessive myogenic noise—were marked and excluded from further analysis.

Following filtering, the continuous data were segmented into 1.5-second epochs time-locked to the marker placed by the participant immediately before each utterance. Each epoch comprised 500 ms preceding the marker (representing the period of attentional focus and preparation for the speech command) and 1000 ms following the marker (covering the execution phase—either overt articulation or subvocal rehearsal—and a brief return to baseline). This temporal structure was chosen to capture both the cognitive preparation and the motor implementation of the speech task, including early and late neurophysiological components relevant for distinguishing between overt and covert conditions.

Potential epoch overlap was considered negligible, as the mean inter-trial interval substantially exceeded the epoch duration of 1.5 seconds across all participants and blocks.

The resulting data matrices, of dimensions [channels × time points] for each session, were concatenated into a final tensor of size [session × channels × time points]. To remove artifacts from the EEG signals, we applied independent component analysis (ICA)²⁰ using the Infomax algorithm as implemented in MNE-Python. Prior to ICA, the data dimensionality was reduced via PCA, with the number of components determined by the rank of the data. Components were inspected manually and excluded if they exhibited topographies and time courses characteristic of ocular artifacts (eye blinks and eye movements). For several participants, components reflecting cardiac artifacts were additionally removed. Typically, 1–3 components were removed per participant.

Data Records

All data files are publicly available on Zenodo²¹. The dataset is organised into three top-level directories, as illustrated in Fig. 4. The raw/ directory contains continuous EEG/EMG recordings in EDF format, organised by language group (russian/, spanish/) and then by participant (sub1/–sub12/ for Russian-speaking and sub0/–sub9/ for Spanish-speaking participants). The preprocessed/ directory follows the same structure and contains epoched data in FIF format and event files in XLSX format for each participant. The metadata/ directory contains the participant metadata file subject_metadata.json. Three additional files are provided at the top level: README.md with dataset documentation, manifest.csv listing all files, and data_dictionary.md describing all variables and fields.

Each.edf file contains a continuous EEG recording (38 channels) acquired over the entire experimental session. The average recording duration is 70 minutes. The.edf format includes all EEG/EMG data, event markers, and supplementary metadata such as sampling rate, channel names, applied filters, and other relevant information. The raw recordings corresponding to events contain time-locked markers synchronized with the recorded signals. Each event code, its identifier (ID), and its description are listed in Table 2.

The EEG data for each participant (i.e., per session), preprocessed according to the pipeline described above, are stored in.fif files. Each file contains an MNE “Epochs” object that encapsulates all trials from the corresponding session. The resulting data tensor has dimensions [trials × 38 × 750]. The number of trials varies across participants and sessions, as detailed in Tables 3, 4 below. The number of channels is 38, and the number of time points is 750, corresponding to a 1.5-second epoch sampled at a final rate of 500 Hz.

Each event file (in.xlsx format) contains a three-column matrix, with each row representing a single trial. The first column contains the event timestamp in samples, the second column indicates the presence of a marker (flag), and the third column specifies the session number.

The metadata file “subject_metadata.json” contains information about participants and the corresponding experimental session details.

Technical Validation

Power spectral analysis. Spectral analysis is one of the most widely used approaches for EEG analysis, enabling decomposition of a complex time-series signal into its constituent frequency components²². It is employed to assess the brain’s functional state and can be effectively used to validate experimental conditions, including comparisons between rest, overt speech, and inner speech. Spectral power (SPM) represents a measure of the average signal power at each point along the frequency axis and is typically expressed in units of squared

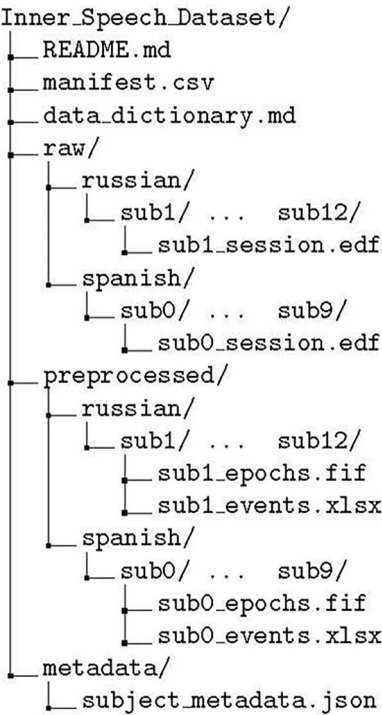


Fig. 4 Database structure.

Label	ID (Russian)	ID (Spanish)	English	Russian	Spanish	Type
BACK1	1	1	back	назад	atrás	overt
BACK2	2	2				covert
DOWN1	3	3	down	вниз	abajo	overt
DOWN2	4	4				covert
FORWARD1	5	5	forward	вперёд	adelante	overt
FORWARD2	6	6				covert
GO	7	7	rest			service: rest
GZ	8	8	relax			service: relax
LEFT1	9	9	left	влево	izquierda	overt
LEFT2	10	10				covert
NEXT1	11	—	next	далее	—	overt (Russian only)
NEXT2	12	—			—	covert (Russian only)
RIGHT1	13	11	right	вправо	derecha	overt
RIGHT2	14	12				covert
UP1	15	13	up	вверх	arriba	overt
UP2	16	14				covert

Table 2. Description of event ID. Note: Spanish-language participants did not include the “next” command. Consequently, event IDs for RIGHT, UP differ between Russian and Spanish datasets. Users should refer to the ID column corresponding to the participant’s language group.

amplitude per hertz ($\mu V^2/Hz$). Given that the boundaries of EEG frequency bands reported in the literature vary across sources, it is essential to explicitly define them in each study (Table 5)²³. Spectral power values are displayed in Fig. 5.

Considering the non-stationary nature of the raw EEG signal, SPM estimates were computed using Welch’s method—averaging periodograms across multiple non-overlapping epochs of the original signal—to enhance statistical reliability and reduce the influence of random fluctuations. Each epoch had a duration of 1.5 seconds. Using this approach, SPM metrics were calculated for 5-Hz-wide frequency bins within the bands specified in Table 5. A one-way analysis of variance (ANOVA) was performed on these SPM measures across three conditions: baseline (“eyes open”), overt speech, and inner speech. All ANOVA analyses reported here should be regarded as exploratory, as no correction for multiple comparisons was applied across channels and frequency bands. The results were stored in the database under the “ANOVA results” folder. For illustrative purposes, the

Subject	Language	Total trials	Mean per word	Min per word	Max per word
sub2	Russian	1208	86	82	104
sub4		1219	87	29	132
sub6		358	26	20	30
sub7		1306	93	85	137
sub8		1237	88	81	108
sub9		1286	92	80	124
sub11		1179	84	81	91
sub12		442	32	26	43
sub0	Spanish	1969	164	96	255
sub1		1571	131	107	168
sub2		2148	179	148	226
sub3		1041	87	61	106
sub4		1108	92	70	110
sub5		1597	133	73	185
sub6		1492	124	106	160
sub7		1560	130	113	154
sub8		1365	114	90	141
sub9		1207	101	79	127

Table 3. Standard marker protocol.

Subject	Language	Total epochs	Marker protocol
sub1	Russian	684	Single marker per utterance series; epochs at 1-sec intervals
sub3		672	Single marker per utterance series; epochs at 1-sec intervals
sub5		684	Single marker per utterance series; epochs at 1-sec intervals
sub10		672	Single marker per utterance series; epochs at 1-sec intervals

Table 4. Modified marker protocol.

Rhythm	Frequency bands (Hz)
Delta (δ)	0.5–4 Hz
Theta (θ)	4–8 Hz
Alpha (α)	8–13 Hz
Beta-1 (β_1)	13–18 Hz
	18–20 Hz
Beta-2 (β_2)	20–25 Hz
	25–30 Hz
Gamma (γ)	30–35 Hz
	35–40 Hz
	40–45 Hz
	45–50 Hz
	50–55 Hz
	55–60 Hz
	60–65 Hz
	65–70 Hz

Table 5. Frequency bands.

55–65 Hz frequency range is highlighted in Table 6 as showing numerically pronounced differences across conditions. These analyses should be considered exploratory and interpreted with caution, as this frequency range may be susceptible to residual myogenic artifacts, and no correction for multiple comparisons was applied. Additional one-way ANOVAs were also conducted to compare SPM measures between different spoken words and between overt versus covert articulation (Tables 7–9).

To enable a comparative analysis of spatially specific EEG patterns recorded during rest, overt speech, and inner speech, topographic maps of spectral power were constructed for the major frequency bands (delta, theta, alpha, beta-1, beta-2). For a more detailed characterization of regional differences, we employed t-distributed stochastic neighbor embedding (t-SNE)²⁴, a nonlinear dimensionality reduction technique. For each session and each of the 38 EEG channels, the average power spectral density (PSD) was computed within predefined

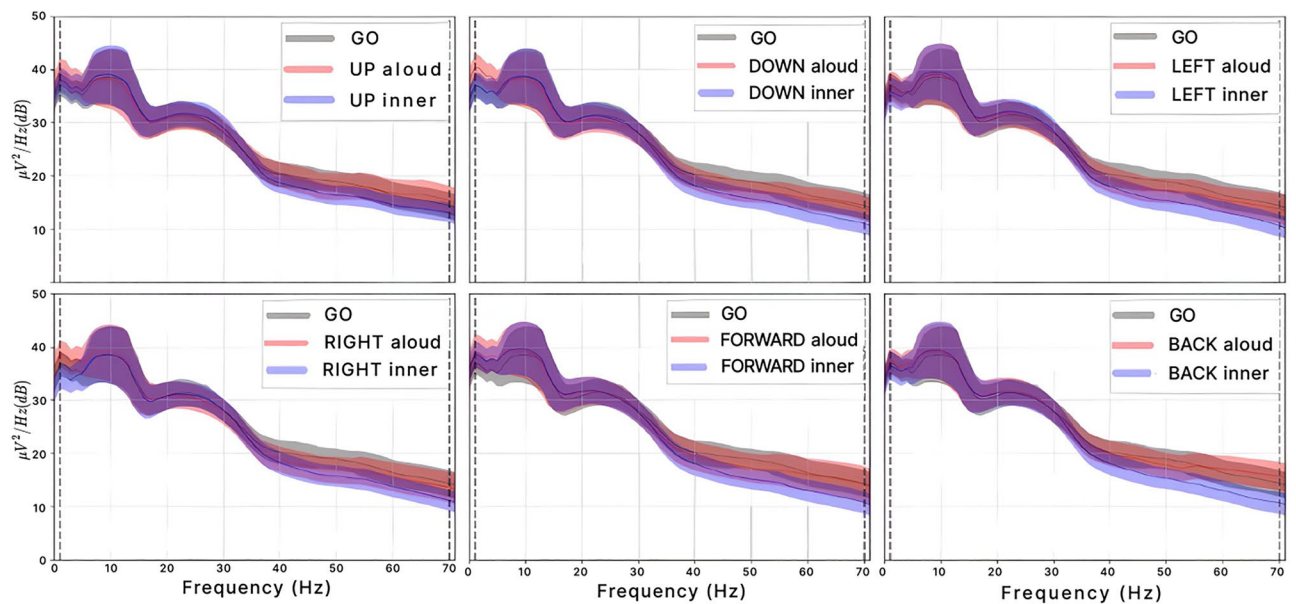


Fig. 5 Average power spectra of the compared states (“eyes open” (GO), saying the word aloud and mentally). Vertical dashed axes on the graph highlight the frequency range 1–70 Hz, in which the band-pass filter was applied.

	Up	Down	Left	Right	Forward	Back
Fp1	0,041	0,039	0,026	0,034	0,01	0,084
Fp2	0,033	0,109	0,031	0,023	0,012	0,027
F7	0,075	0,035	0,028	0,21	0,025	0,018
F3	0,066	0,254	0,286	0,029	0,033	0,117
F4	0,069	0,38	0,268	0,173	0,062	0,225
F8	0,093	0,019	0,027	0,021	0,011	0,001
Ft7	0,003	0,038	0,014	0,008	0,064	0,001
Fc3	0,087	0,534	0,156	0,019	0,115	0,074
Fc4	0,055	0,179	0,281	0,23	0,063	0,053
Ft8	0,01	0,02	0,079	0,01	0,006	0,002
T3	0,004	0,037	0,015	0,014	0,071	0,003
C3	0,202	0,334	0,263	0,233	0,117	0,05
C4	0,124	0,049	0,05	0,138	0,125	0,07
T4	0,02	0,01	0,006	0,026	0,017	0,069
Tp7	0,012	0,033	0,054	0,048	0,009	0,071
Cp3	0,022	0,028	0,021	0,042	0,011	0,008
Cp4	0,021	0,006	0,005	0,032	0,019	0,008
Tp8	0,028	0,053	0,004	0,051	0,018	0,006
T5	0,027	0,018	0,014	0,061	0,013	0,032
P5	0,023	0,026	0,02	0,035	0,023	0,002
P3	0,025	0,028	0,031	0,065	0,013	0,027
P4	0,024	0,049	0,054	0,013	0,019	0,003
P6	0,029	0,052	0,011	0,023	0,028	0,027
T6	0,03	0,027	0,012	0,053	0,016	0,004
Po7	0,028	0,035	0,017	0,013	0,025	0,004
Po3	0,008	0,024	0,025	0,017	0,017	0,051
Po4	0,034	0,052	0,031	0,037	0,019	0,02
Po8	0,024	0,033	0,03	0,015	0,01	0,011
O1	0,07	0,411	0,016	0,084	0,172	0,054
O2	0,228	0,455	0,134	0,338	0,212	0,013

Table 6. p-values from ANOVA analysis with words and background state. Cells highlighted in blue contain a p-value less than 0.05. The channels with the greatest difference from the background state are highlighted in bold.

Word Pairs	F7	FT7	T3	T5	P5	P3
Up - Down	0,0312	0,0201	0,0488	0,0354	0,0245	0,0410
Up - Right	0,0471	0,0299	0,0385	0,0225	0,0415	0,0498
Up - Left	0,0255	0,0303	0,0701	0,0440	0,0211	0,0333
Up - Forward	0,0401	0,0289	0,0315	0,0450	0,0366	0,0279
Up - Back	0,0350	0,0490	0,0205	0,0233	0,0419	0,0209
Down - Right	0,0515	0,0645	0,0422	0,1550	0,0390	0,0461
Down - Left	0,0495	0,0260	0,0290	0,0405	0,0435	0,0230
Down - Forward	0,0300	0,0430	0,0375	0,0240	0,0491	0,0285
Down - Back	0,0270	0,0480	0,0210	0,0325	0,0380	0,0475
Right - Left	0,0550	0,0220	0,0444	0,0305	0,0455	0,0265
Right - Forward	0,0249	0,0465	0,0268	0,0880	0,0425	0,0355
Right - Back	0,0370	0,0400	0,0330	0,0470	0,0250	0,0395
Left - Forward	0,0460	0,0219	0,0459	0,0275	0,0630	0,0490
Left - Back	0,0280	0,0439	0,0235	0,0360	0,0200	0,0445
Forward - Back	0,0399	0,0208	0,0480	0,0222	0,0340	0,0295

Table 7. p-values from ANOVA analysis with pair of aloud words. Cells highlighted in blue contain a p-value less than 0.05.

Word Pairs	F7	FT7	T3	T5	P5	P3
Up - Down	0.033	0.049	0.007	0.003	0.046	0.105
Up - Right	0.003	0.015	0.027	0.042	0.019	0.008
Up - Left	0.048	0.001	0.011	0.035	0.004	0.022
Up - Forward	0.021	0.033	0.045	0.006	0.038	0.009
Up - Back	0.067	0.049	0.002	0.125	0.031	0.044
Down - Right	0.036	0.018	0.005	0.029	0.041	0.013
Down - Left	0.010	0.040	0.020	0.003	0.016	0.037
Down - Forward	0.043	0.006	0.034	0.014	0.066	0.001
Down - Back	0.025	0.039	0.009	0.005	0.028	0.048
Right - Left	0.002	0.012	0.047	0.030	0.007	0.024
Right - Forward	0.032	0.045	0.008	0.017	0.049	0.004
Right - Back	0.016	0.023	0.038	0.041	0.001	0.036
Left - Forward	0.040	0.062	0.004	0.026	0.032	0.015
Left - Back	0.005	0.010	0.019	0.002	0.040	0.007
Forward - Back	0.029	0.043	0.031	0.047	0.088	0.020

Table 8. p-values from ANOVA analysis with pair of inner words. Cells highlighted in blue contain a p-value less than 0.05.

sub-bands over the 1.5-second active task window. The collection of these vectors across all sessions—separately for Russian- and Spanish-speaking participants—and across conditions (rest, overt speech, inner speech) formed the input dataset for t-SNE.

The t-SNE algorithm was applied to this vector set to visualize the underlying cluster structure in a two-dimensional space²⁴. Parameters were optimized to maximize cluster separation (perplexity = 30, number of iterations = 1000, learning rate = 200). The t-SNE analysis focused specifically on the 55–65 Hz frequency window. For illustrative purposes, the t-SNE visualization suggests a tendency towards separation between points corresponding to different experimental conditions (Fig. 6). The analysis focused on channels overlying Broca's and Wernicke's areas, where the largest visual separation was observed. It should be noted that t-SNE results are sensitive to parameter choices and should not be interpreted as confirmation of distinct neural activity patterns; they are provided here as an exploratory visualization tool only.

Furthermore, analysis of spatial profiles associated with different words demonstrated that the topographic distribution of activity enabled clear clustering—and thus reliable discrimination—between overt and covert articulation trials.

Thus, t-SNE was applied as a complementary exploratory visualisation to illustrate the spatial distribution of spectral power across conditions. The results are intended to provide an initial qualitative overview and should be interpreted accordingly.

Channel coherence analysis. As is well established, coherence analysis enables the assessment of spatial consistency—and thus functional interdependence and coordination—between neural activities recorded from distinct brain regions²⁵. Coherence measures quantitatively characterize the degree of synchronization between

Word Pairs	F7	Ft7	T3	T5	P5	P3
Up	0.010	0.060	0.020	0.003	0.016	0.037
Down	0.011	0.045	0.003	0.033	0.076	0.049
Left	0.036	0.002	0.029	0.005	0.042	0.018
Right	0.007	0.031	0.048	0.019	0.004	0.035
Forward	0.041	0.014	0.006	0.047	0.025	0.080
Back	0.022	0.040	0.067	0.017	0.038	0.046

Table 9. p-values from ANOVA analysis with pair of aloud and inner words. Cells highlighted in blue contain a p-value less than 0.05.

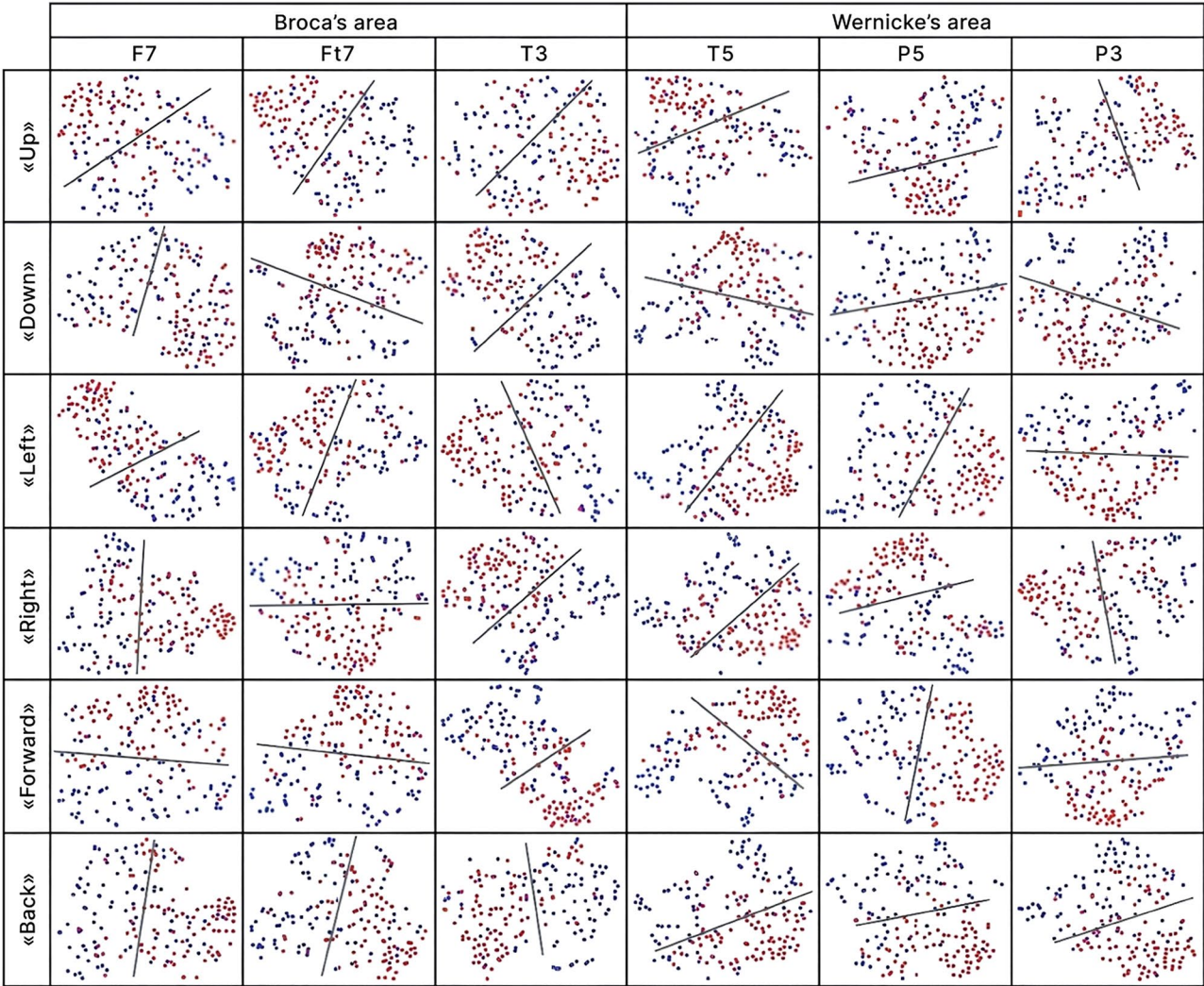


Fig. 6 Scatter plots obtained by the t-SNE method in the 55–65 Hz sub-band for all. direction words in Broca's and Wernicke's areas. The red dots in the diagram correspond to the background state “eyes open”, and the blue dots to the inner speech. The gray dividing lines show a variant of visual separation of states into 2 clusters.

specific frequency components of signals recorded from different electrode pairs. Coherence analysis thereby permits inferences regarding which neocortical regions exhibit connectivity differences related to overt versus covert speech, and within which EEG frequency bands these differences occur.

Coherence metrics were computed for the electrophysiological recordings contained in the described database by calculating coherence coefficients for every possible pair of EEG channels during two conditions: baseline (“eyes open”) and inner speech production. The analysis was carried out across all frequency bands specified in Table 5. Comparative analysis of the resulting coherence values revealed that, during inner speech (relative to baseline), coherence decreased for a subset of electrode pairs, indicating reduced functional coupling—or

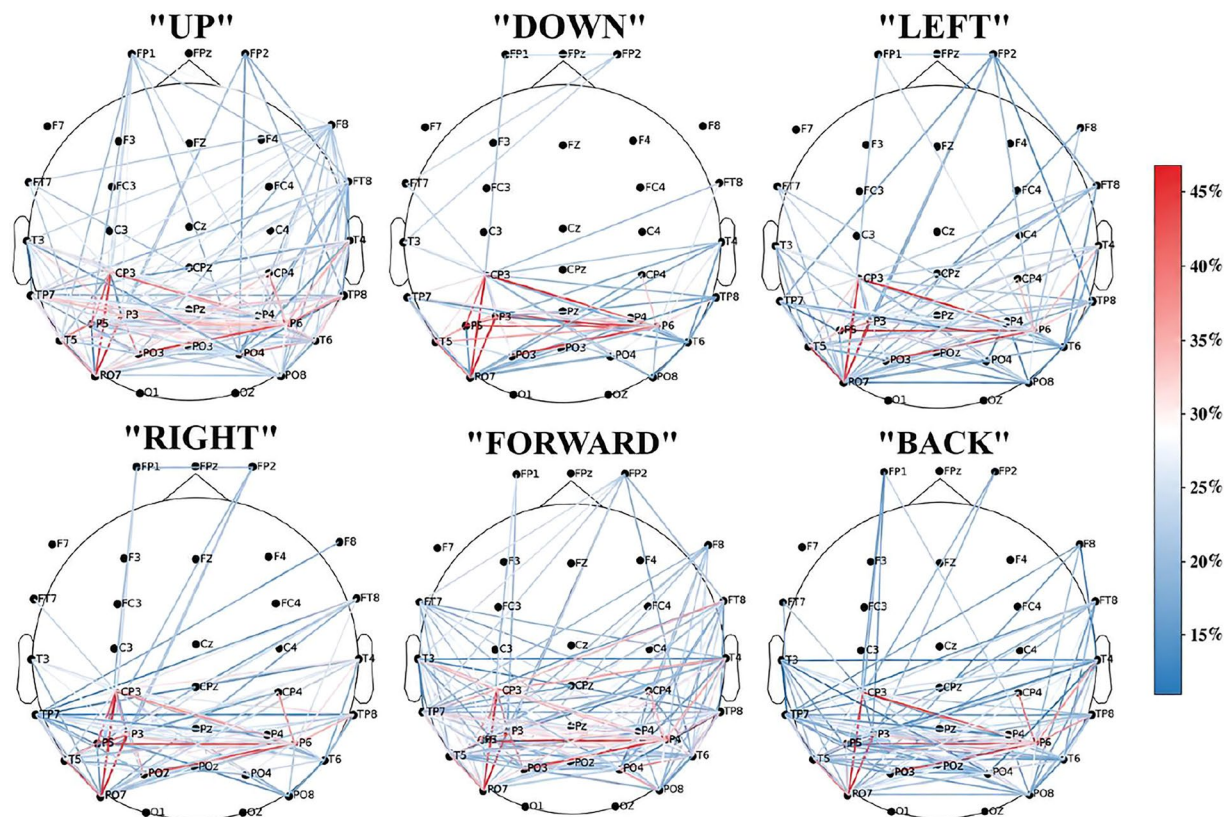


Fig. 7 Comparative coherence analysis of the states of inner speech with the “eyes open” state in the form of topographic maps. The lines connecting the channels show the pairs of channels in which there is a desynchronization during inner speech. The scale on the graph shows by how many percent the coherence of the background state exceeds the coherence of the inner speech.

desynchronization—between the corresponding cortical regions. The electrode pairs contributing most strongly to the discrimination between inner speech and rest were again those overlying Broca’s and Wernicke’s areas (see Fig. 7). This finding aligns with the neurolinguistic model of speech, which posits that these regions play a central role in the generation and comprehension of verbal information^{26,27}.

Classification. Standard machine learning classifiers — Random Forest, Support Vector Machine (SVM) with a linear kernel, and Linear Discriminant Analysis (LDA) — were applied to spectral power (SPM) and coherence features computed within the 55–65 Hz frequency band as a sanity check of data quality. Classification was performed per participant using 5-fold cross-validation. The highest accuracy for discriminating between the three conditions (rest, overt speech, and inner speech) was achieved using Random Forest ($78 \pm 4\%$), followed by SVM ($74 \pm 5\%$) and LDA ($67 \pm 6\%$). These results suggest that the recorded signals contain condition-specific information detectable by standard classifiers, supporting the basic quality of the dataset. These analyses should be regarded as exploratory; a more thorough benchmarking is left to future users of the dataset. The use of these approaches for EEG-based classification is well established^{28,29} and has been explored by our group in prior work on inner speech EEG classification^{30–33}.

Usage Notes

Scripts for downloading the dataset, loading EDF recordings, and reproducing epoched data are available in the accompanying code repository (see Code Availability). The directory structure, file formats, and event marker descriptions are provided in Fig. 4 and Tables 2–4.

Audio was not recorded during the experimental sessions; speech timing and word duration information is therefore not available in the dataset.

Users should note the following when working with the data. First, event marker IDs differ between Russian-speaking and Spanish-speaking participants — refer to Table 2 for language-specific ID mappings. Second, four Russian-speaking participants (sub1, sub3, sub5, sub10) used a modified marker protocol with a different epoch structure; these are described in Table 4 and should be treated separately in trial-level analyses. Third, event markers were self-initiated by a right-hand keypress, and the associated motor artifact may partially overlap with the early post-marker window (0–200 ms). Fourth, Russian and Spanish stimulus words differ in syllable count and phonological structure (Table 1), which should be considered when making cross-linguistic signal-level comparisons. Fifth, stimulus words were presented in a predominantly fixed order across participants, which may introduce order-related confounds including fatigue effects towards the end of the session.

Data availability

The dataset is publicly available on Zenodo²¹. The repository contains raw EEG recordings in.edf format, preprocessed epoched data in.fif format, event files in.xlsx format, and participant metadata in subject_metadata.json.

Code availability

Custom scripts for downloading the dataset, loading EDF recordings, and reproducing epoched data are available at <https://github.com/AvedikEkiz/inner-speech-bci>. All preprocessing was performed using standard functions from MNE-Python and standard Python scientific libraries (NumPy, SciPy, Pandas).

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Author contributions

D.K. performed statistical analysis and wrote the original draft. P.S. contributed to conceptualization and supervised the study. A.E. performed data curation, data processing, and database organization. I.S. developed experimental methodology and conducted the experiments. D.S. performed final article revision and project administration. I.V. provided a report on technical validation analysis of data. V.K. contributed to methodology development and performed final article revision. All authors reviewed and approved the final manuscript.

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Competing interests

The authors declare no competing interests.

Additional information

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